26CA1103E - LINUX SYSTEMS PROGRAMMING

MINI PROJECT REPORT

ON

PERSONAL FILE ORGANIZER



Team Name: THE NEXT LEVEL

Submitted by

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Department	CSA – BCA
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Course Coordinator	Dr. Alampally Sreedevi
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CERTIFICATE

This is to certify that the mini project titled "**Student Marks & Grade Calculator**" has been successfully completed by BOMMIDENI POSHIKA (R.No:2620520011), SRIRANGAM NIKITHA (R.No:2610520027), **J SHOWRYA KIRAN**(R.No:2610520029) **and POOJA** (R.No:2610520044) of Team THE NEXT LEVEL. I Year – I Semester, Department of CSA – BCA, KL University, as part of the **26CA1103E -LINUX SYSTEMS PROGRAMMING** course during the academic year 2026–2027.

The project demonstrates the use of Linux command-line tools and Bash shell scripting to solve a simple academic problem

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Dr. Alampally Sreedevi **Course Coordinator**	**Dr. D. S. Rao** **HOD- CSA-BCA**

DECLARATION

We hereby declare that the project report titled "**Student Marks & Grade Calculator**" is our own work, carried out as part of the 26CA1103E -LINUX SYSTEMS PROGRAMMING **course** in the Department of CSA – BCA, KL University. We have used Bash shell scripting and standard Linux commands to design and implement the project. The project has been prepared for academic learning and demonstration purposes.

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Date: 12/10/2026
Place: Hyderabad

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We also thank the Head of the Department **Dr. D.S.Rao**, and our institution for providing the Linux / WSL environment required for implementation.

This project helped us understand basic Linux commands, Bash scripting, variables, conditions, arithmetic operations, functions, input/output, and file handling. Finally, we thank our friends and family for their support.

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ABSTRACT

The Student Marks & Grade Calculator is a beginner-friendly Linux Bash project designed to automate the calculation of a student's total marks, average, grade, and result status. The application accepts marks for multiple subjects, validates the basic input, calculates the total and average, and assigns a grade using predefined conditions.

The project runs entirely from the Linux terminal and does not require a graphical interface or database. It introduces students to practical shell scripting concepts such as variables, read commands, arithmetic expressions, if-else statements, functions, and formatted output.

Keywords: *Linux, Bash, Shell Scripting, Marks Calculator, Grade, Terminal*

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1. INTRODUCTION

Linux provides a powerful command-line environment in which users can automate repetitive tasks using shell scripts. Bash is one of the most widely used shells in Linux. A small file-management application is a useful way for beginners to learn how commands and programming logic work together.

This project uses Bash to organize files in a directory by their extensions. The user enters a source directory, and the script creates a folder for each file type and moves the matching files into it. Files without extensions are placed in a “No Extension” folder.

2. PROBLEM STATEMENT

Create a beginner-friendly Bash script named organize.sh using the nano text editor. The script should display a heading, create folders for documents and images, and create a sample file in each folder. It should then print a clearly formatted list of three file names in the order they are created, leave a blank line, and display a separate WORK FILES section containing three work file names. The script should use simple commands such as echo, mkdir, and touch. After writing the script, save it, run it in the terminal, and confirm that the folders and files are created and the file names appear in the expected order.

3. OBJECTIVES

- Create and run a simple Bash script using the nano text editor.
- Use touch to create sample files in the Documents and Images folders.
- Use echo to display the project title and file names in order.
- Use a blank line to separate the file list from the WORK FILES section.
- Practice basic Linux command-line skills
- Understand the Linux terminal and basic Bash scripting.

4. SCOPE OF THE PROJECT

This project covers creating and running a simple Bash script named organize.sh using the nano text editor. The script displays the PERSONAL FILE ORGANIZER heading, create Documents/notes.txt and Images/photo.jpg, and prints the specified file names under the WORK FILES heading.

The project is limited to practicing basic Linux commands such as touch and echo. It does not sort, move, or copy existing files. Platform: Linux operating system Interface: Command-line terminal using the Bash shell and nano text editor Input: The user enters commands in the terminal and runs the organize.sh script.

The script uses predefined file names. Output: The script displays the project heading and file names in order. It creates notes.txt and photo.jpg in the specified folders. Users: Students or faculty demonstrating basic shell scripting and file handling.

5. EXISTING SYSTEM

The files are managed manually through the Linux terminal. The user must enter commands individually to create files and display their names. This can take more time and may lead to typing mistakes. There is no script to present the file names in a consistent order.

6. PROPOSED SYSTEM

The proposed system uses a simple Bash script named organize.sh, written in the nano text editor. When run, it displays the PERSONAL FILE ORGANIZER heading, creates Documents/notes.txt and Images/photo.jpg using touch, and displays the listed file names and WORK FILES section using echo. The script automates these specific steps, but it does not sort, move, or copy existing files.

7. REQUIREMENTS

7.1 Hardware Requirements

- Computer or laptop.
- Minimum 4 GB RAM recommended.
- Keyboard and display.

7.2 Software Requirements

- Ubuntu Linux or Ubuntu through Windows Subsystem for Linux (WSL).
- Bash shell.
- Terminal application.
- Any text editor such as nano or vim.

8. SYSTEM DESIGN

1. The user opens a terminal and creates or edits organize.sh using nano.
2. The user runs the script with bash organize.sh.
3. The script displays PERSONAL FILE ORGANIZER.
4. The touch commands create Documents/notes.txt and Images/photo.jpg.
5. The echo commands display the file names, a blank line, and the WORK FILES section. Commands used:

- echo — displays headings and file names.
- touch — creates empty files.
- nano — edits the script.
- bash — runs the script.

9. PROCESS FLOW

Start

↓

Open the Linux terminal

↓

Create or edit organize.sh using nano

↓

Enter the Bash commands

↓

Save the script and exit nano

↓

Run: bash organize.sh

↓

Display “PERSONAL FILE ORGANIZER”

↓

Create Documents/notes.txt and Images/photo.jpg

↓

Display the file names in order

↓

Display “WORK FILES” and its file names

↓

End

10. ALGORITHM

1. Start.

2. Open the terminal.

3. Create or open organize.sh using nano.

4. Enter the Bash script.

5. Display PERSONAL FILE ORGANIZER.

6. Create Documents/notes.txt and Images/photo.jpg using touch.

7. Display File1.txt, File2.txt, and File3.txt in order.

8. Print a blank line.

9. Display WORK FILES.

10. Display Workfile 1, Workfile 2, and Workfile 3 in order.

11. Save the script and exit nano.

12. Run the script with bash organize.sh.

13. Stop.

11. EXECUTION

```
poshika@LAPTOP-N4HEEPQC:~$ mkdir desktop
poshika@LAPTOP-N4HEEPQC:~$ cd desktop
poshika@LAPTOP-N4HEEPQC:~/desktop$ nano organize.sh
```

poshika@LAPTOP-N4HEEPQC

```
  GNU nano 8.7.1                        organize.sh
#!/bin/bash

echo "PERSONAL FILE ORGANIZER"

touch Documents/notes.txt
touch Images/photo.jpg

echo "File1.txt"
echo "File2.txt"
echo "File3.txt"
echo "           "
echo "WORK FILES"
echo "Workfile 1"
echo "Workfile 2"
echo "Workfile 3"
```

```
poshika@LAPTOP-N4HEEPQC:~/desktop$ bash organize.sh
PERSONAL FILE ORGANIZER
File1.txt
File2.txt
File3.txt

WORK FILES
Workfile 1
Workfile 2
Workfile 3
```

12. Team Members and Responsibilities

TEAM MEMBER	RESPONSIBILITIES	DELIVERABLES
POSHIKA BOMMIDENI	Problem analysis, path validation, file operations Bash logic	Source code and algorithm
NIKITHA SRIRANGAM	Testing, sample outputs, documentation, presentation	Project report
J SHOWRYA KIRAN	Research	DATA
POOJA	Research	DATA

13. ADVANTAGES

- Easy for beginners to understand and use.
- Automates the display of headings and file names.
- Creates the specified empty files using touch.
- Runs from the Linux terminal with a simple command.

14. LIMITATIONS

- File names are fixed in the script and cannot be entered by the user.
- It only creates the files; it does not organize existing files.
- The Documents and Images folders must already exist, or touch may fail to create the files.
- Running the script repeatedly displays the same names and leaves existing files unchanged.

15. FUTURE ENHANCEMENTS

- Add commands to create the Documents and Images folders automatically.
- Let users enter their own folder and file names.
- Sort existing files into folders by type.
- Check whether a file already exists before creating it.
- Display a message confirming each file was created.

16. LEARNING OUTCOMES

- Learned how to create and run a Bash script on Linux.
- Practiced using nano to write and save a script.
- Learned how echo displays text and touch creates empty files.
- Understood how a script runs commands in sequence. Store student records in a text or CSV file.

17. CONCLUSION

The Personal File Organizer project demonstrates basic Bash scripting. It displays headings and file names and creates sample files in specified folders. While it does not organize existing files, it provides a foundation for learning how to automate file-related tasks in Linux.

18. REFERENCES

- The Linux man-pages: bash, mkdir and related commands.
- GNU Bash Reference Manual.
- Ubuntu Linux documentation.
- Course notes and laboratory material provided by the faculty.